Module 2

1. Fraud Transaction Detection (Linear Search)

```
import java.util.Scanner;

Public class FraudTransactionDetection {
    Public static void main (String[] args) {
        Scanner sc = new Scanner (system.in);

        int n = sc.nextInt ();
        int [] transactions = new int [n];
        }
        int suspiciousId = sc.next Int ();
        boolean found = false;

        for (int i=0; i<n ; i++) {

         if (transaction[i] == suspicious Id ) {
             found = true;
             break;
         }
      }
        if (found) {
          System.out.println ("Fraud Transaction
                                     Found");
        }
        else {
          System.out.println ("Transaction Not
                                     Found");
        }
          sc.close ();
    }
}
```

~~Output~~. Input —

5

1201 1305 1450 1408 1502

1450.

Output.

Fraud Transaction found

Another Input.

5

1201 1305 1450 1408 1502

Output.

Transaction not found.

(2) Server Log Search (Binary Search)

```
import java.util.Scanner;

Public class ServerLogSearch {
public static void main(string[] args){
    Scanner sc = new Scanner (system.in);

    int n = sc.nextInt();
    int[] logs = new int[n];
    for (int i=0; i<n; i++) {
        log[i] = sc.nextInt();
    }
    int target = sc.nextInt();
    int left = 0, right = n-1;
    boolean found = false;

    while (left <= right) {
        int mid = left + (right - left) / 2;
        if (logs[mid] == target) {
            found = true;
```

```
        break;
    } else if (logs[mid] < target) {
        left = mid + 1;
    } else {
        right = mid - 1;
    }
}
    if (found) {
        System.out.println("Log Found");
    } else {
        System.out.println("Log Not Found");
    }
        sc.close();
    }
}
```

Input:

7

1001 1005 1010 1015 1020 1030 1040

1015

Output

Log Found

Another Input:

7

1001 1005 1010 1015 1020 1030 1040

1025

Output

Log Not Found.

8. Sort Employee Salaries (Bubble Sort)

```
import java.util.Scanner;
public class EmployeeSalarySort {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int[] salaries = new int[n];
        for (int i=0; i<n; i++) {
            salaries[i] = sc.nextInt();
        }
        for (int i = 0; i<n-1; i++) {
            for (int j = 0; j<n-i-1; j++) {
                if (salaries[j] > salaries[j+1]) {
                    int temp = salaries[j];
                    salaries[j] = salaries[j];
                    salaries[j] = salaries[j+1];
                    salaries[j+1] = temp;
                }
            }
        }
        for (int i=0; i<n; i++) {
            System.out.println(salaries[i]+" ");
        }
        sc.close();
    }
}
```

Input
5
45000 32000 55000 28000 60000

Output
28000 32000 45000 55000 60000

4. Sort Website Response Time (Insertion Sort)

```
import java.util.scanner;
public class WebsiteResponseTimeSort {
    public static void main (String[] args){
        Scanner sc = new Scanner (system.in);
        int n = sc.nextInt ();
        int [] responseTimes = new int [n];
        for (int i=0; i<n; i++) {
            responseTimes[i] = sc.nextInt ();
        }
        for (int i=1; i<n ;i++){
            int key = responseTimes [i];
            int j = i-1;
        while (j>=0 && responseTimes[j] >key){
        responseTimes [j+1] = responseTimes [j];
            j--;
        }
        responseTimes [j+1]= key;
    }
    for(int i= 0; i<n; i++){
      system.out.println (responseTime[i]+" ")
    }
      sc.close ();
   }
}
```

Input
5
250 120 320 180 100
Output
100 120 180 250 320.

Prioritize Bug Report (selection sort)

```
import java.util.scanner;
public class BugReportPrioritySort {
    public static void main(string[] args){
        Scanner sc = new Scanner(system.in);
        int n = sc.nextInt();
        int[] priorities = new int[n];
        for (int i = 0; i<n ; i++) {
            priorities[i] = sc.nextInt();
        }
        for (int i = 0; i<n-1; i++) {
            int minIndex = i;
        for (int j = i+1; j<n ; j++) {
            if (priorities [j] < priorities (minIndex]) {
            minIndex = j;
        }
    }
        int temp = priorities[i];
        priorities [i] = priorities [minIndex];
        priorities [minIndex] = temp;
    }
        for (int i=0; i<n ; i++) {
        System.out.println (priorities[i] + " ");
        }
        sc.close();
    }
}
```

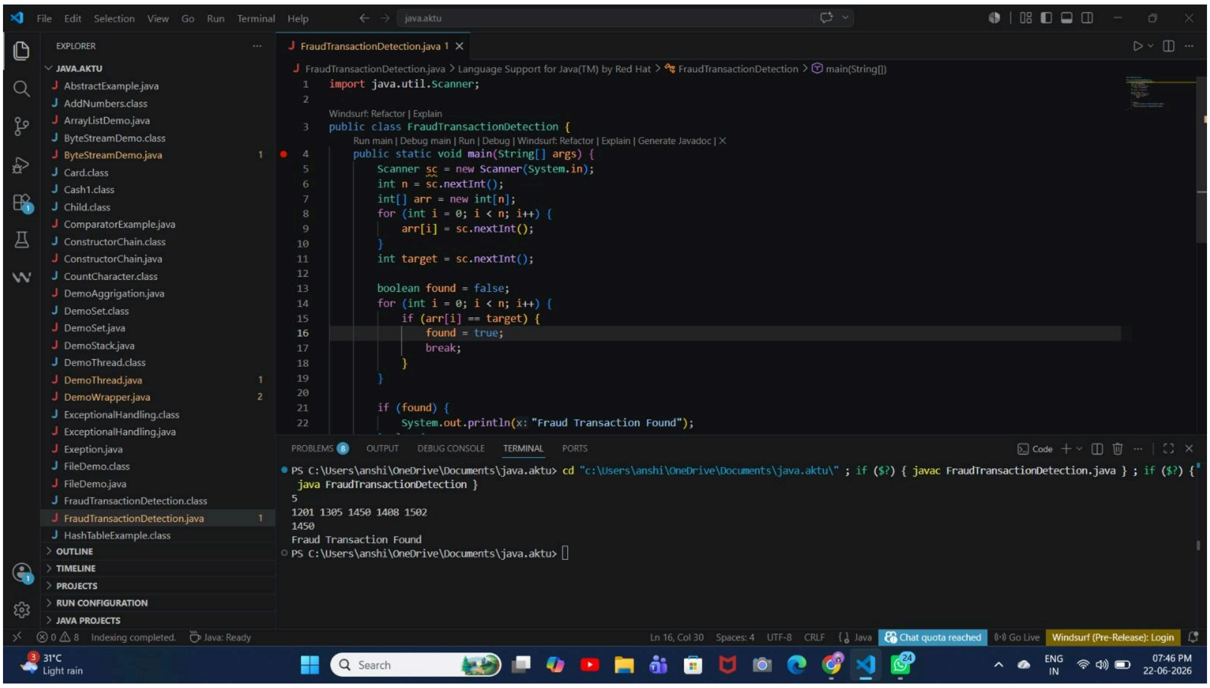
Input

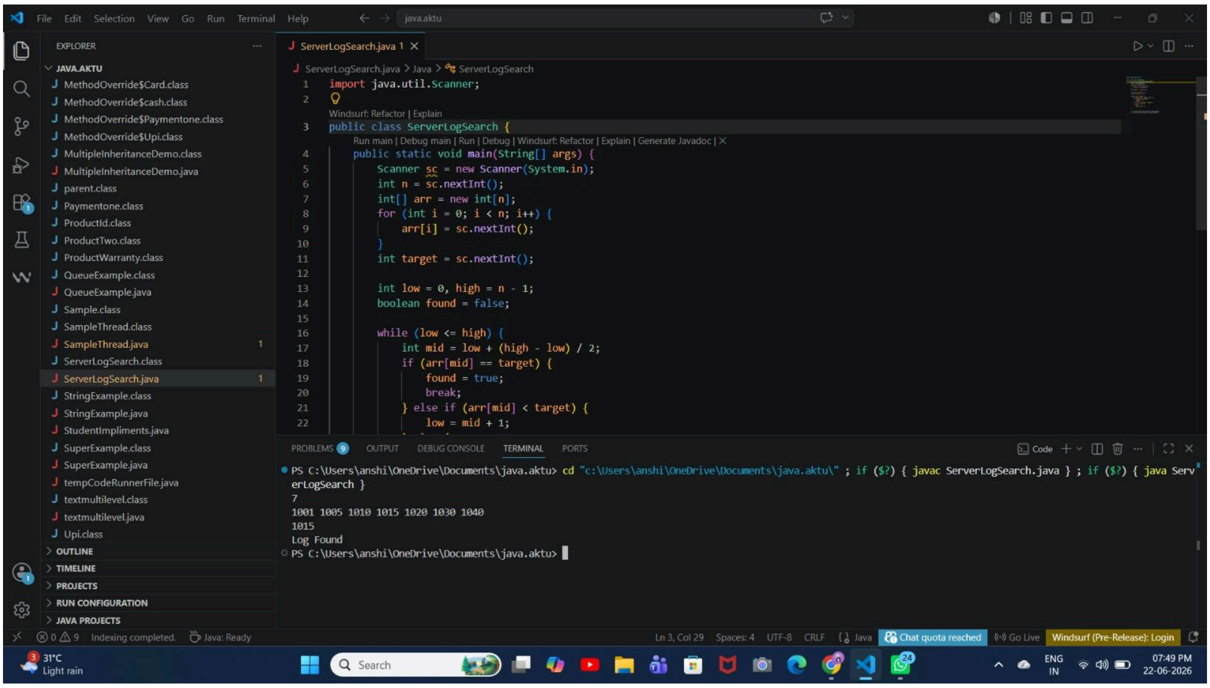
5

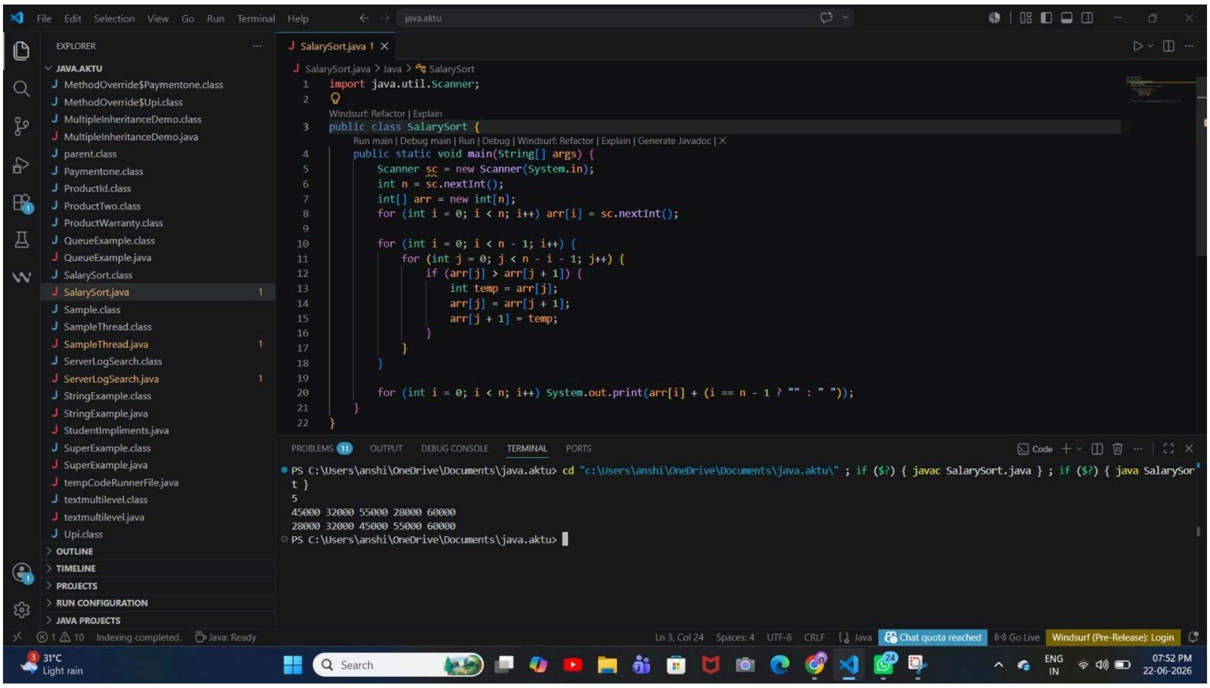
9 2 8 1 5

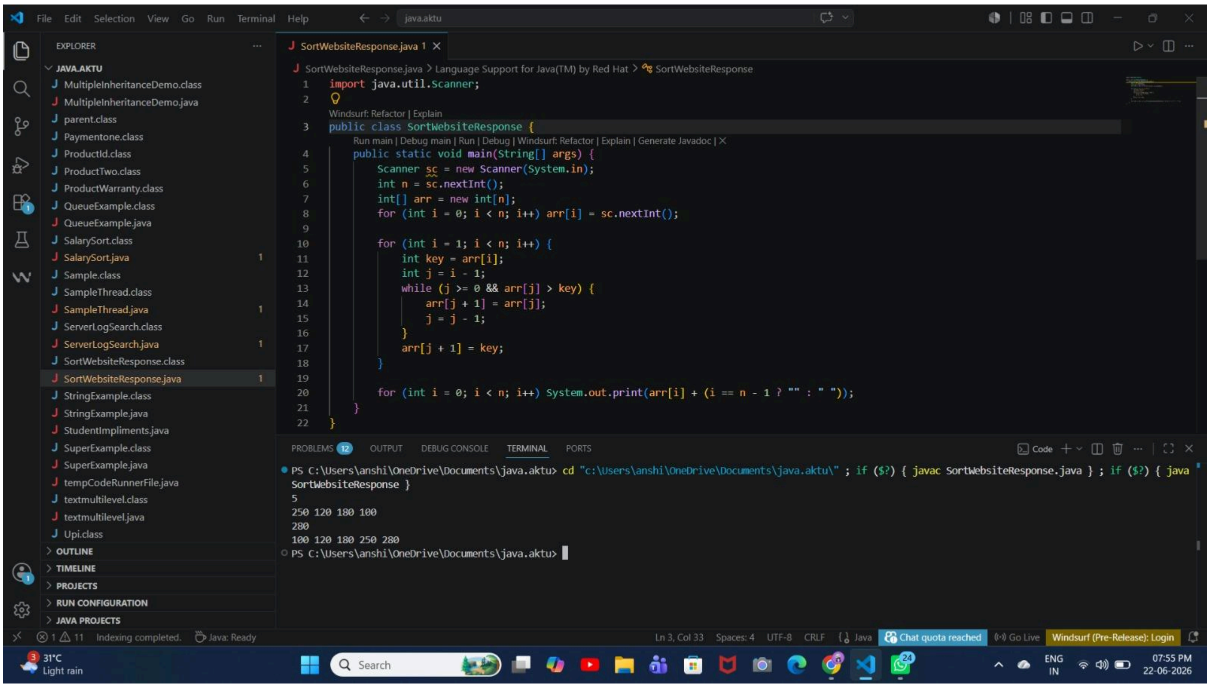
Output :

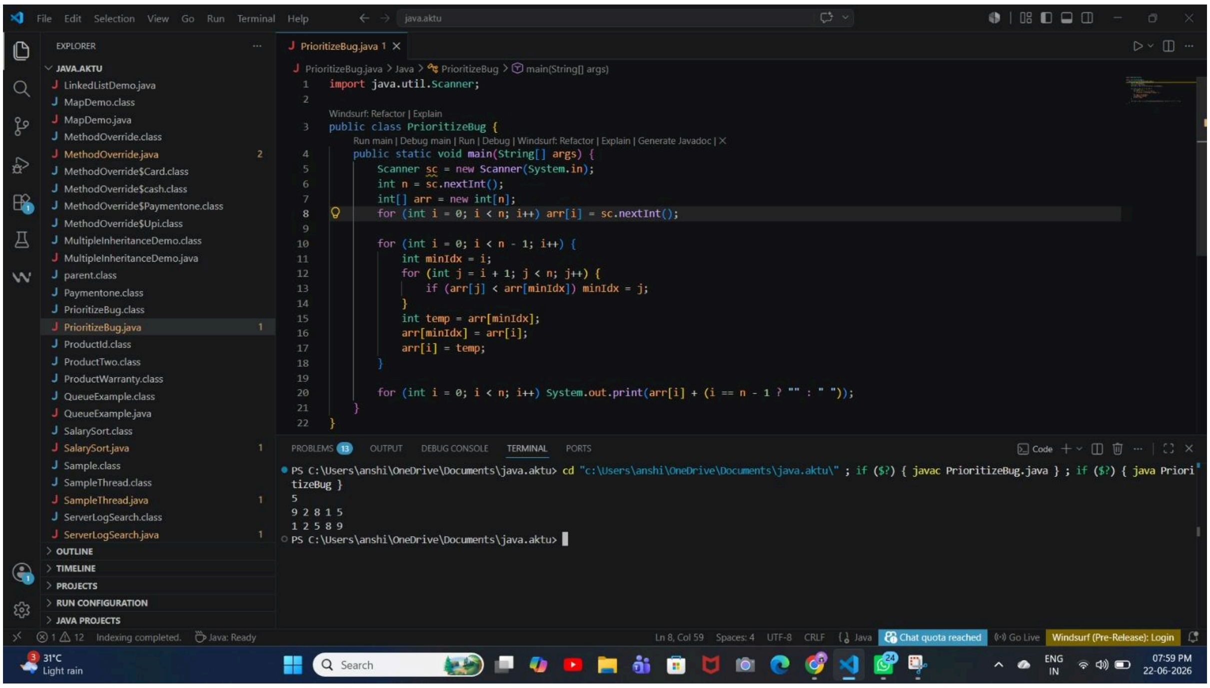
1 2 5 8 9


FraudTransactionDetection.java
import java.util.Scanner;
public class FraudTransactionDetection {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int n = sc.nextInt();
int[] arr = new int[n];
for (int i = 0; i < n; i++) {
arr[i] = sc.nextInt();
}
int target = sc.nextInt();
boolean found = false;
for (int i = 0; i < n; i++) {
if (arr[i] == target) {
found = true;
break;
}
}
if (found) {
System.out.println(x: "Fraud Transaction Found");
PS C:\Users\anshi\OneDrive\Documents\java.aktu> cd "c:\Users\anshi\OneDrive\Documents\java.aktu\" ; if ($?) { javac FraudTransactionDetection.java } ; if ($?) { java FraudTransactionDetection }
5
1201 1305 1450 1408 1502
1450
Fraud Transaction Found
PS C:\Users\anshi\OneDrive\Documents\java.aktu>


ServerLogSearch.java
import java.util.Scanner;
public class ServerLogSearch {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int n = sc.nextInt();
int[] arr = new int[n];
for (int i = 0; i < n; i++) {
arr[i] = sc.nextInt();
}
int target = sc.nextInt();
int low = 0, high = n - 1;
boolean found = false;
while (low <= high) {
int mid = low + (high - low) / 2;
if (arr[mid] == target) {
found = true;
break;
} else if (arr[mid] < target) {
low = mid + 1;
PS C:\Users\anshi\OneDrive\Documents\java.aktu> cd "c:\Users\anshi\OneDrive\Documents\java.aktu\" ; if ($?) { javac ServerLogSearch.java } ; if ($?) { java ServerLogSearch }
7
1001 1005 1010 1015 1020 1030 1040
1015
Log Found
PS C:\Users\anshi\OneDrive\Documents\java.aktu>


File Edit Selection View Go Run Terminal Help
java.aktu
EXPLORER
JAVA.AKTU
MethodOverride$Paymentone.class
MethodOverride$Upi.class
MultipleInheritanceDemo.class
MultipleInheritanceDemo.java
parent.class
Paymentone.class
ProductId.class
ProductTwo.class
ProductWarranty.class
QueueExample.class
QueueExample.java
SalarySort.class
SalarySort.java
Sample.class
SampleThread.class
SampleThread.java
ServerLogSearch.class
ServerLogSearch.java
StringExample.class
StringExample.java
StudentImpliments.java
SuperExample.class
SuperExample.java
tempCodeRunnerFile.java
textmultilevel.class
textmultilevel.java
Upi.class
OUTLINE
TIMELINE
PROJECTS
RUN CONFIGURATION
JAVA PROJECTS
SalarySort.java 1
SalarySort.java > Java > SalarySort
import java.util.Scanner;
Windsurf: Refactor | Explain
public class SalarySort {
Run main | Debug main | Run | Debug | Windsurf: Refactor | Explain | Generate Javadoc |
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int n = sc.nextInt();
int[] arr = new int[n];
for (int i = 0; i < n; i++) arr[i] = sc.nextInt();
for (int i = 0; i < n - 1; i++) {
for (int j = 0; j < n - i - 1; j++) {
if (arr[j] > arr[j + 1]) {
int temp = arr[j];
arr[j] = arr[j + 1];
arr[j + 1] = temp;
}
}
}
for (int i = 0; i < n; i++) System.out.print(arr[i] + (i == n - 1 ? "" : " "));
}
}
PROBLEMS 11
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS
Code
PS C:\Users\anshi\OneDrive\Documents\java.aktu> cd "c:\Users\anshi\OneDrive\Documents\java.aktu\" ; if ($?) { javac SalarySort.java } ; if ($?) { java SalarySort }
5
45000 32000 55000 28000 60000
28000 32000 45000 55000 60000
PS C:\Users\anshi\OneDrive\Documents\java.aktu>
Indexing completed.
Java: Ready
Ln 3, Col 24
Spaces: 4
UTF-8
CRLF
Java
Chat quota reached
Go Live
Windsurf (Pre-Release): Login
31°C
Light rain
Search
ENG
IN
07:52 PM
22-06-2026


SortWebsiteResponse.java
import java.util.Scanner;
public class SortWebsiteResponse {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int n = sc.nextInt();
int[] arr = new int[n];
for (int i = 0; i < n; i++) arr[i] = sc.nextInt();
for (int i = 1; i < n; i++) {
int key = arr[i];
int j = i - 1;
while (j >= 0 && arr[j] > key) {
arr[j + 1] = arr[j];
j = j - 1;
}
arr[j + 1] = key;
}
for (int i = 0; i < n; i++) System.out.print(arr[i] + (i == n - 1 ? "" : " "));
}
}
PS C:\Users\anshi\OneDrive\Documents\java.aktu> cd "c:\Users\anshi\OneDrive\Documents\java.aktu\" ; if ($?) { javac SortWebsiteResponse.java } ; if ($?) { java SortWebsiteResponse }
5
250 120 180 100
280
100 120 180 250 280
PS C:\Users\anshi\OneDrive\Documents\java.aktu>


PrioritizeBug.java
import java.util.Scanner;
public class PrioritizeBug {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int n = sc.nextInt();
int[] arr = new int[n];
for (int i = 0; i < n; i++) arr[i] = sc.nextInt();
for (int i = 0; i < n - 1; i++) {
int minIdx = i;
for (int j = i + 1; j < n; j++) {
if (arr[j] < arr[minIdx]) minIdx = j;
}
int temp = arr[minIdx];
arr[minIdx] = arr[i];
arr[i] = temp;
}
for (int i = 0; i < n; i++) System.out.print(arr[i] + (i == n - 1 ? "" : " "));
}
}
PS C:\Users\anshi\OneDrive\Documents\java.aktu> cd "c:\Users\anshi\OneDrive\Documents\java.aktu\" ; if ($?) { javac PrioritizeBug.java } ; if ($?) { java PrioritizeBug }
5
9 2 8 1 5
1 2 5 8 9
PS C:\Users\anshi\OneDrive\Documents\java.aktu>